SET 25: WEBSITE TRAFFIC ANALYSIS

Dataset	Website Traffic
Department	AIML
Course Code / Course Name	DSA0602 / Data Handling and Visualization
Experiment	Data Visualization using R
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Day	Tuesday

Dataset: Website Traffic

Date	Page Views	Click-through Rate
2023-01-01	1500	2.3%
2023-01-02	1600	2.7%
2023-01-03	1400	2.0%
2023-01-04	1650	2.4%
2023-01-05	1800	2.6%

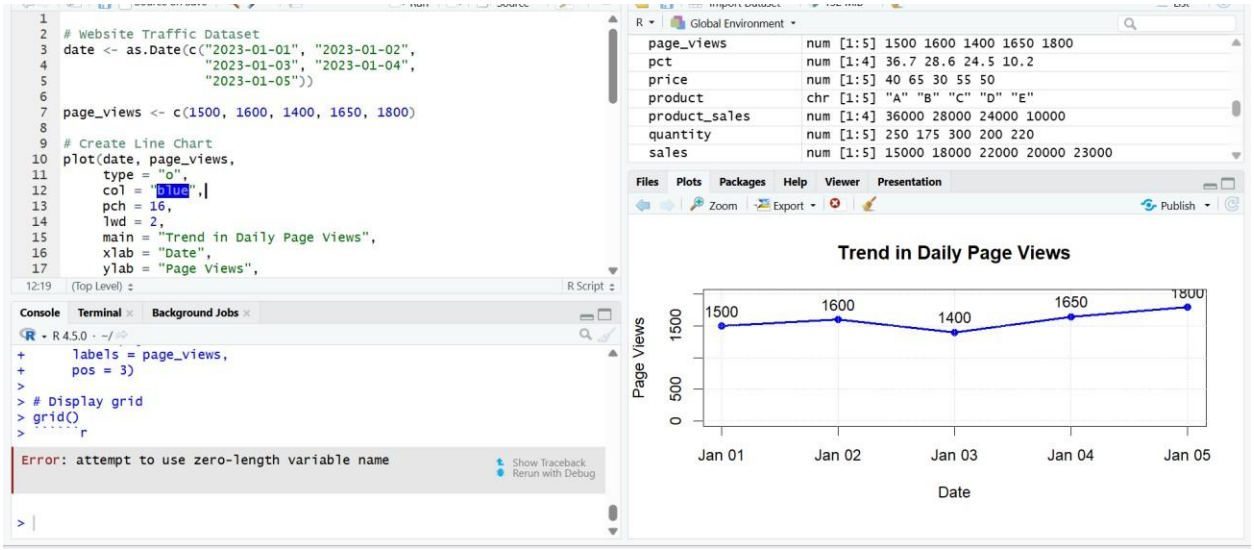
PROGRAM 1: Tableau Line Chart of Page Views

Question: In Tableau, create a line chart to visualize the trend in page views over time. Label the axes and title the chart.

Tableau Steps

1. Open Tableau and connect to the Website Traffic dataset (Excel/CSV).
2. Verify the fields Date, Page Views and Click-through Rate; make sure Date is a Date type and Page Views is a number.
3. Create a worksheet named “Page Views Trend”. Drag Date to Columns (set to Exact Date) and Page Views to Rows.
4. Select Line in the Marks card.
5. Add the title “Trend in Daily Page Views”, label the axes (Date, Page Views) and show mark labels.

Output 1: Daily Page Views Line Chart (Tableau)



Observation: Page views rose from 1,500 to 1,600, dipped to 1,400 on 2023-01-03, then climbed to 1,650 and reached the peak of 1,800 on 2023-01-05. Overall the trend is upward.

PROGRAM 2: Bar Chart of Top N Days by Click-through Rate

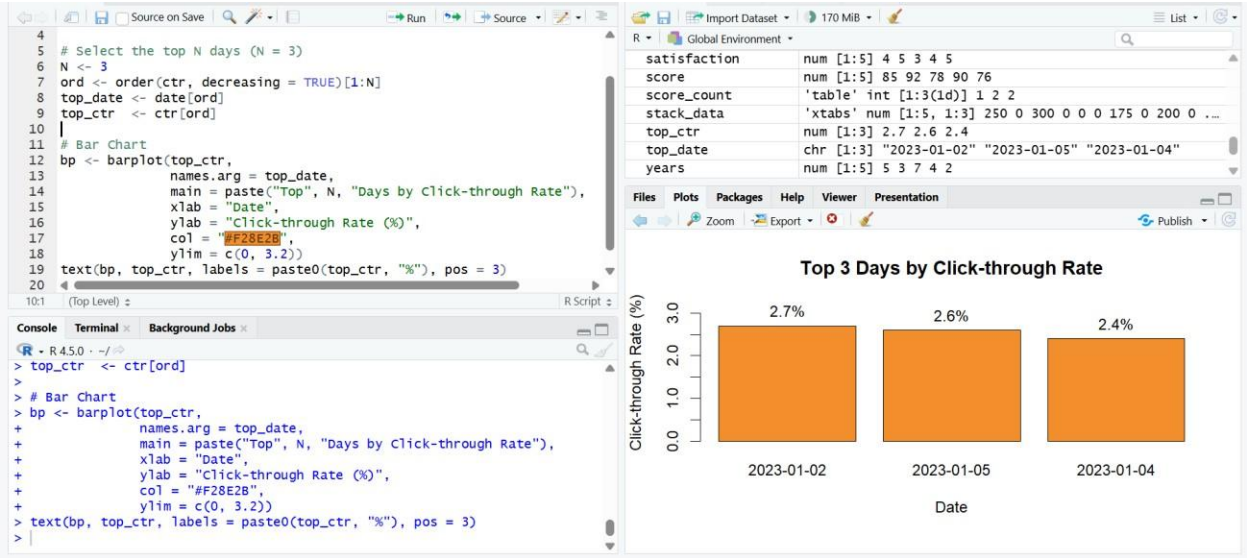
Question: In R, generate a bar chart to show the top N days with the highest click-through rates. Label the chart elements.

```
# Dataset
date <- c("2023-01-01", "2023-01-02", "2023-01-03", "2023-01-04", "2023-01-
05")
ctr  <- c(2.3, 2.7, 2.0, 2.4, 2.6)

# Select the top N days (N = 3)
N <- 3
ord <- order(ctr, decreasing = TRUE)[1:N]
top_date <- date[ord]
top_ctr  <- ctr[ord]

# Bar Chart
bp <- barplot(top_ctr,
              names.arg = top_date,
              main = paste("Top", N, "Days by Click-through Rate"),
              xlab = "Date",
              ylab = "Click-through Rate (%)",
              col = "#F28E2B",
              ylim = c(0, 3.2))
text(bp, top_ctr, labels = paste0(top_ctr, "%"), pos = 3)
```

Output 2: Top Days by Click-through Rate Bar Chart



Observation: With N = 3, the top days are 2023-01-02 (2.7%), 2023-01-05 (2.6%) and 2023-01-04 (2.4%). The days 2023-01-01 (2.3%) and 2023-01-03 (2.0%) are left out. The click-through rate was lowest on the day with the fewest page views.

PROGRAM 3: Stacked Area Chart of User Interactions

Question: In R, build a stacked area chart to display the distribution of user interactions (likes, shares, comments) on the website.

Note: this data is not given in the question, so sample values are assumed for the demonstration. Replace them with your actual data if you have it.

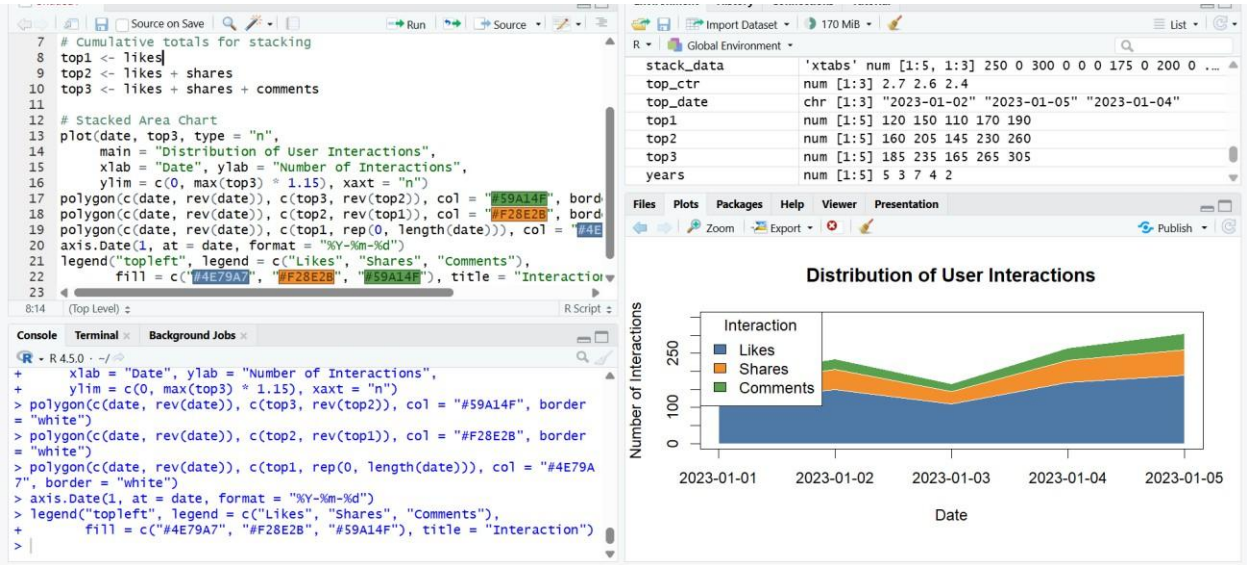
Date	Likes	Shares	Comments
2023-01-01	120	40	25
2023-01-02	150	55	30
2023-01-03	110	35	20
2023-01-04	170	60	35
2023-01-05	190	70	45

```
# Dataset (interaction counts are assumed sample data)
date     <- as.Date(c("2023-01-01", "2023-01-02", "2023-01-03", "2023-01-04",
"2023-01-05"))
likes    <- c(120, 150, 110, 170, 190)
shares   <- c(40, 55, 35, 60, 70)
comments <- c(25, 30, 20, 35, 45)

# Cumulative totals for stacking
top1 <- likes
top2 <- likes + shares
top3 <- likes + shares + comments
```

```
# Stacked Area Chart
plot(date, top3, type = "n",
     main = "Distribution of User Interactions",
     xlab = "Date", ylab = "Number of Interactions",
     ylim = c(0, max(top3) * 1.15), xaxt = "n")
polygon(c(date, rev(date)), c(top3, rev(top2)), col = "#59A14F", border =
"white")
polygon(c(date, rev(date)), c(top2, rev(top1)), col = "#F28E2B", border =
"white")
polygon(c(date, rev(date)), c(top1, rep(0, length(date))), col = "#4E79A7",
border = "white")
axis.Date(1, at = date, format = "%Y-%m-%d")
legend("topleft", legend = c("Likes", "Shares", "Comments"),
       fill = c("#4E79A7", "#F28E2B", "#59A14F"), title = "Interaction")
```

Output 3: User Interactions Stacked Area Chart



Observation: Likes form the largest layer on every day (about 64% of all interactions), followed by shares (about 23%) and comments (about 13%). Total interactions fell to the lowest value (165) on 2023-01-03 and peaked at 305 on 2023-01-05, following the same pattern as the page views.

PROGRAM 4: Tableau Dashboard: Traffic Source Map and Page Views by Source

Question: In Tableau, create a dashboard with an interactive map showing traffic sources and a bar chart displaying page views by source.

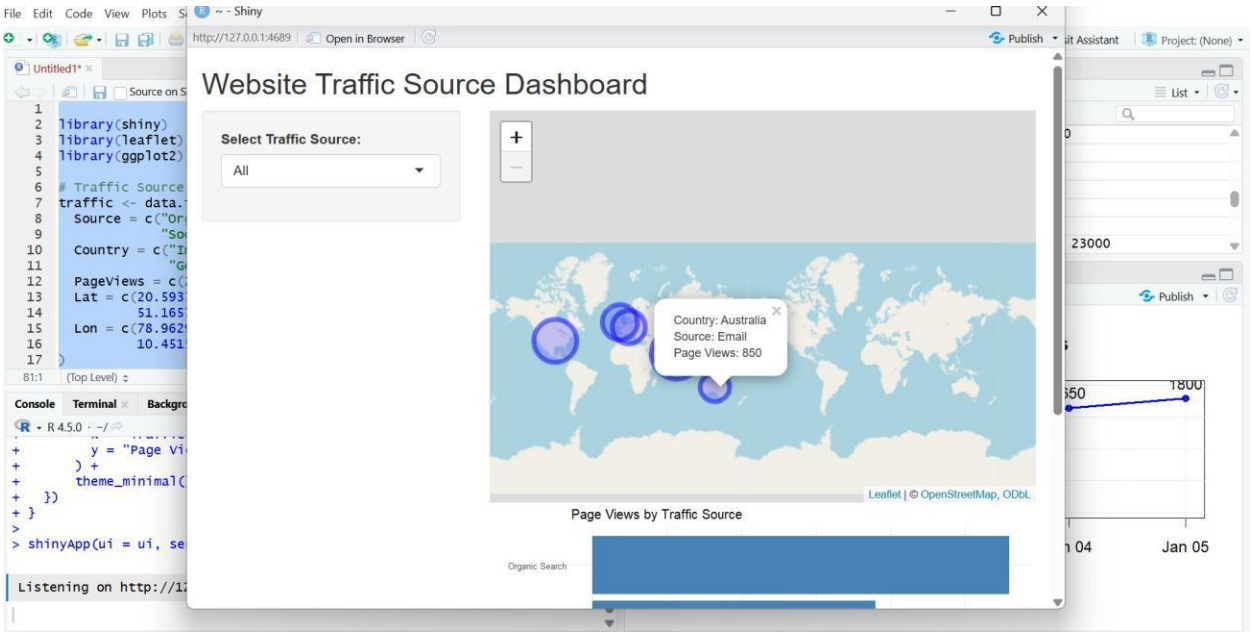
Note: traffic source data is not given in the question, so sample values are assumed (they add up to the same total of 7,950 page views as the daily data).

Traffic Source	Country	Page Views
Organic Search	India	2800
Direct	USA	1900
Social Media	UK	1300
Referral	Germany	1100
Email	Australia	850

Tableau Steps

1. Open Tableau and connect to the traffic source data (Excel/CSV) with the fields Traffic Source, Country and Page Views.
2. Check that Country has the geographic role Country/Region (a globe icon appears next to it); set it manually if needed.
3. Create a worksheet named “Traffic Source Map”. Double-click Country to create the map, drag Traffic Source to Color and Page Views to Size.
4. Add the title “Traffic Sources by Country”, show the Traffic Source legend and add Page Views to Tooltip.
5. Create a worksheet named “Page Views by Source”. Drag Traffic Source to Columns and Page Views to Rows; select Bar in the Marks card.
6. Sort the bars in descending order, show mark labels and add the title “Page Views by Traffic Source”.
7. Create a new Dashboard, set the size to Automatic, and drag in the map and the bar chart.
8. Add the title “Website Traffic Source Dashboard”, click the “Use as Filter” icon on the map and add a Traffic Source quick filter (Show Filter).
9. Test the dashboard by clicking a country on the map or a bar and checking that the other view updates.

Output 4: Traffic Source Dashboard (Tableau)



Observation: Organic Search brings the most page views (2,800, about 35% of the total) and Email the fewest (850, about 11%). Clicking a country on the map filters the bar chart to its traffic source, and the quick filter allows exploration by source.

Procedure

1. Open Tableau and import the Website Traffic dataset; create the line chart of page views.
2. In R, sort the click-through rates with order(), select the top N days and draw the bar chart using barplot().
3. Create the likes, shares and comments data, calculate cumulative totals and draw the stacked area chart using polygon().
4. Import the traffic source data into Tableau and create the map and the bar chart as separate worksheets.
5. Combine the worksheets into a dashboard, add a title, and enable filters.
6. Test the dashboard interactivity.

Result

The Website Traffic data was successfully visualized using a Tableau line chart of page views, a bar chart of the top days by click-through rate, a stacked area chart of user interactions, and an interactive Tableau dashboard with a traffic source map and a page views by source bar chart.